

Fixed-Discriminant Gauss Reduction: Cycles, Stabilizers, Reversal, and a Finite Determinant

HCS-C364 source-local theorem package

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Abstract

At one positive nonsquare binary-form discriminant we freeze an exact primitive reduced phase space and prove that the Gauss shift is its permutation. Every orbit has a primitive continued-fraction word whose positive matrix product is a hyperbolic stabilizer; the same matrix gives an exact cycle multiplier and its discriminant identity. Conjugate reciprocity reverses the dynamics, while the complete cycle inventory yields every fixed-point count, the finite Artin–Mazur zeta, and the Koopman determinant. Finite exact computations audit conventions only; the theorem is analytic.

1 The convention and the finite phase space

Fix a positive nonsquare integer $\Delta \equiv 0, 1 \pmod{4}$. Let $\mathcal{R}_\Delta$ contain the numbers

$$\alpha = \frac{P + \sqrt{\Delta}}{Q}, \quad Q = 2A > 0, \quad A = \frac{Q}{2}, \quad B = -P, \quad C = \frac{P^2 - \Delta}{2Q}, \quad (1)$$

for which $A, B, C \in \mathbb{Z}$, $\gcd(A, B, C) = 1$, and

$$\alpha > 1, \quad -1 < \bar{\alpha} = \frac{P - \sqrt{\Delta}}{Q} < 0. \quad (2)$$

The polynomial $Ax^2 + Bx + C$ has discriminant exactly Δ and roots $(P \pm \sqrt{\Delta})/Q$. Thus (1) is a binary-form discriminant convention, not a radicand convention that accidentally changes the discriminant to 4Δ .

Define

$$G(\alpha) = \frac{1}{\alpha - \lfloor \alpha \rfloor}. \quad (3)$$

Theorem 1 (Complete reduced dynamics). *The set $\mathcal{R}_\Delta$ is finite and G is a bijection. Let $\alpha_0, \dots, \alpha_{\ell-1}$ be a minimal cycle and $a_j = \lfloor \alpha_j \rfloor$. Let $A(a)$ denote the two-by-two matrix with rows $(a, 1)$ and $(1, 0)$. Then*

$$M = A(a_0)A(a_1) \cdots A(a_{\ell-1}). \quad (4)$$

Writing $M_{11} = a, M_{12} = b, M_{21} = c, M_{22} = d$, this matrix fixes α_0 , has $\det M = (-1)^\ell$, and is the primitive positive inverse-branch stabilizer. For some integer $k > 0$,

$$(c, d - a, -b) = k(A, B, C), \quad (\operatorname{tr} M)^2 - 4 \det M = k^2 \Delta. \quad (5)$$

If $\lambda = c\alpha_0 + d$, then $\lambda > 1$ and

$$|(G^\ell)'(\alpha_0)| = \lambda^2. \quad (6)$$

Proof. From (2), $|P| < \sqrt{\Delta}$ and $0 < Q < P + \sqrt{\Delta} < 2\sqrt{\Delta}$, proving finiteness. Put $a = \lfloor \alpha \rfloor$,

$$P_1 = aQ - P, \quad Q_1 = \frac{\Delta - P_1^2}{Q}. \quad (7)$$

Rationalization gives $G(\alpha) = (P_1 + \sqrt{\Delta})/Q_1$. Because $0 < \alpha - a < 1$, this root exceeds one; because $-a - 1 < \bar{\alpha} - a < -a$, its conjugate lies in $(-1, 0)$. The original integrality says $2Q \mid P^2 - \Delta$, while $P_1^2 - P^2 = a^2Q^2 - 2aPQ$ is divisible by $2Q$; hence

Q_1 is positive and even and the new form is integral. The inverse branch $x \mapsto a + 1/x$ is unimodular, so form content, and hence primitivity, is preserved.

Conversely, for reduced β , set

$$t = -1/\bar{\beta}, \quad a = \lfloor t \rfloor, \quad \alpha = a + 1/\beta. \quad (8)$$

Here $t > 1$ is irrational, so $\alpha > 1$ and $\bar{\alpha} = a - t \in (-1, 0)$. The inverse unimodular substitution preserves the integral primitive form and $G(\alpha) = \beta$. The conjugate interval determines a uniquely, proving bijectivity.

Writing $H_a(x) = a + 1/x$, one has $\alpha_j = H_{a_j}(\alpha_{j+1})$; therefore (4) fixes α_0 . A shorter digit word would, by uniqueness of an infinite simple continued fraction, give a shorter state return. Each factor has determinant minus one. The integral stabilizer of a real quadratic irrational maps through $N \mapsto c_N \alpha + d_N$, after choosing the positive representative, to units of its quadratic order that are positive at the α -embedding. We use the classical reduction–automorph cross-section lemma in its exact return form: an expanding positive projective stabilizer preserves the oriented geodesic from $\bar{\alpha}$ to α and advances its reduced cross-section by an integer $r \geq 1$ of complete Gauss digits. Thus

$$[N] = [A(a_0) \cdots A(a_{r-1})], \quad G^r(\alpha_0) = \alpha_0,$$

and every complete return gives such a stabilizer. This standard continued-fraction correspondence is source-anchored to Uspensky and Buell below. Minimality of ℓ makes every $r = n\ell$, and periodicity makes the corresponding product M^n ; a representative first written with mixed signs therefore cannot yield a smaller positive return. Hence M generates the positive projective stabilizer. Since $\det(M^n) = (-1)^{\ell n}$, the determinant-one stabilizer up to $\pm I$ is generated by M for even ℓ and by M^2 for odd ℓ .

The fixed-point equation of M is $cx^2 + (d - a)x - b = 0$. Its coefficient vector is an integral multiple of the primitive vector (A, B, C) , giving (5). Finally $M'(\alpha) = \det M / (c\alpha + d)^2$; since M is the local inverse of G^ℓ , reciprocal differentiation gives (6). Positive digit matrices give $\lambda > 1$. This is the *reduced-permutation owner*. QED

2 Reversal and the complete determinant

Define

$$R(\alpha) = -\frac{1}{\bar{\alpha}}, \quad R(P, Q) = (P, (\Delta - P^2)/Q). \quad (9)$$

Proposition 1 (Reversing involution). *The map R preserves $\mathcal{R}_\Delta$, satisfies $R^2 = 1$, and $RGR = G^{-1}$. It sends every cycle word to its reversal.*

Proof. The two inequalities in (2) are exchanged by conjugate reciprocity; rationalization gives the pair formula in (9), from which $R^2 = 1$. Since the digit matrices are symmetric, transposing $A(a_0) \cdots A(a_{\ell-1})$ reverses its word. Equivalently,

$$[\overline{a_0, \dots, a_{\ell-1}}] \mapsto [\overline{a_{\ell-1}, \dots, a_0}].$$

Reversal conjugates a left shift to a right shift, proving $RGR = G^{-1}$. QED

Let c_m count cycles of length m , and let $U_G f = f \circ G$ on $\ell^2(\mathcal{R}_\Delta)$. A length- m cycle contributes all its states to $\text{Fix}(G^n)$ exactly when $m \mid n$, hence

$$\#\text{Fix}(G^n) = \sum_{m \mid n} m c_m. \quad (10)$$

Substitution into the finite Artin–Mazur definition, followed by the determinant of a cyclic permutation matrix, gives

$$\zeta_{G,\Delta}(z) = \prod_m (1 - z^m)^{-c_m}, \quad \det(I - zU_G) = \prod_m (1 - z^m)^{c_m} = \zeta_{G,\Delta}(z)^{-1}. \quad (11)$$

This is the *reversal-determinant owner*. It is a finite source cycle identity, not an Euler product over rational primes.

3 Exact regression receipt

An independent implementation enumerates primitive form coefficients rather than the producer’s pair loop. For all 469 admissible discriminants through 1000, it checks 5,387 states, 775 complete cycles, 11,256 fixed-power rows, the update and inverse, $RGR = G^{-1}$, every matrix identity in (5), and every factor in (11). The independent checker passes 117,887 assertions; a separate SymPy lane passes 3,403 exact identities; 47 repaired-hash attacks are rejected. Two isolated directories reproduce the same 3,307,442 evidence bytes. These finite facts test conventions; they do not prove the infinite family.

Source and collision boundary

Uspensky's reduction paper appears in the *Bulletin of the AMS* (1930), 710–718, doi:10.1090/S0002-9904-1930-05043-0. Buell's chapter "Indefinite Forms" fixes authoritative form conventions, doi:10.1007/978-1-4612-4542-1_3. Zagier supplies real-quadratic arithmetic context, doi:10.1007/978-3-642-61829-1. These sources establish lineage, not priority for this packaging. C16 uses continued fractions in an S-arithmetic clock, C193 is a Markoff–Vieta descent tree, and C330 is Romik's Pythagorean map; none owns this fixed-discriminant finite reduction permutation.